Smooth Numbers in Short Intervals

Ernest S. Croot III

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Abstract

We show that for any $\epsilon > 0$, there exists a number $c(\epsilon) > 0$, such that, for all x sufficiently large, there are at least $x^{1/2}(\log x)^{-\log 4+o(1)}$ integers $n \in [x, x + c(\epsilon)\sqrt{x}]$, all of whose prime factors are $\leq x^{3/(14\sqrt{e})+\epsilon}$.

1 Introduction.

There are many unsettled questions concerning the distribution of integers having no "large" (the meaning of how large is *large* depends on context) prime factors. Such integers are called "smooth numbers", and we say that an integer n is y-smooth if all its prime divisors are $\leq y$. For an interval $[a, b]$ define $\psi([a, b], y)$ to be the number of y-smooths lying in $[a, b]$. The following is one such important, unsettled question:

Conjecture 1

$$\psi([x, x + \sqrt{x}], x^\theta) > 0, \quad \text{for all } 0 < \theta \leq 1, \text{ and all } x > x_0(\theta).$$

G. Harman [4] has recently shown that this conjecture holds for $y = x^{1/(4\sqrt{e})+\epsilon}$ and any $\epsilon > 0$; moreover, he has shown that there are $\gg \sqrt{x}$ such y-smooths in $[x, x + \sqrt{x}]$, and has obtained results for even shorter intervals, at the expense of a larger smoothness bound. For intervals of length much longer than $\sqrt{x}$, much more is known. For instance, A. Balog showed in [1] that

$$\psi([x, x + x^{1/2+\epsilon}], x^\alpha) \gg_{\epsilon,\alpha} x^{1/2+\epsilon},$$

and many researchers have proved various refinements. For instance, in [3], J. Friedlander and A. Granville showed that the $x^{1/2+\epsilon}$ can be replaced with any

$$z \ > \ y^2\sqrt{x}\exp((\log x)^{1/6}),$$

and x^α with y, where

$$\exp((\log x)^{5/6+o(1)}) \ \leq \ y \ \leq \ x.$$

There are also some conditional improvements; for instance, in [6], Xuan showed that the Riemann Hypothesis implies $[x, x+\sqrt{x}(\log x)^{1+o(1)}]$ contains an x^ϵ-smooth.

In this paper we prove the following

Theorem 1 *For every $\epsilon > 0$, there exists a $c = c(\epsilon) > 0$, such that*

$$\psi([x, x+c\sqrt{x}], x^{3/(14\sqrt{e})+\epsilon}) \ > \ \frac{\sqrt{x}}{\log^{\log 4+o(1)} x},$$

where the implied constant in the $o(1)$ depends on ϵ.

This result improves upon Harman's for intervals of length more than a constant times $\sqrt{x}$; however, Harman's still gives the strongest result for intervals of length exactly $\sqrt{x}$.

In the next section we give the proof of Theorem 1.

2 Proof of Theorem 1.

Fix $0 < \epsilon < 1/200$. Let $y_0 = (1+\epsilon)^{-1}\sqrt{x}$, $y_1 = (1+\epsilon)\sqrt{x}$, $y = y_0$; let N be the set of integers in $[y_0, y_1]$; and finally, let $V = V(B)$ be the set of integers h in N such that $[h+1, h+B]$ contains a $y^{3/(7\sqrt{e})+20\epsilon}$-smooth integer having fewer than $\log\log(y) + (\log\log y)^{2/3}$ prime power divisors. We will show below that

$$|V| \ > \ \left(\frac{1}{2} + 2\epsilon\right)|N|. \tag{1}$$

If we can prove this, then we claim that the main theorem will follow. To see how, let P be the set of all pairs (m, n), where $m \in [y_0, \sqrt{x}]$ is an integer, and where $n = \lfloor x/m \rfloor$. We claim that all but at most $\epsilon|N|$ of the integers h in N occur as a coordinate of one of the pairs in P: First, note that the first coordinates of all the pairs are distinct. The second coordinates are all distinct from the first coordinates since they (second) lie in $[\sqrt{x}, y_1]$. They (second) are distinct from each other since for $m \in [y_0, \sqrt{x}]$ we have x/m and $x/(m-1)$ differ by at least one, which means $\lfloor x/m \rfloor < \lfloor x/(m-1) \rfloor$. In total, the number of integers in N appearing as coordinates of pairs in P is then $2|P| > (1-\epsilon)|N|$.

So, at least $(1/2+\epsilon)|N|$ of the elements $h \in V$ must appear as a coordinate in P. Since there are at most $|N|/2$ pairs in P, this means that there are at least

$\epsilon|N|$ pairs $(m,n) \in P$ such that both $m, n \in V$. For each such pair, let $m_0 \in [m+1, m+B]$ and $n_0 \in [n+1, n+B]$ be $y^{3/(7\sqrt{e})+20\epsilon}$-smooth numbers, each having $< \log\log y + (\log\log y)^{2/3}$ prime power factors. These numbers are clearly $H = x^{3/(14\sqrt{e})+20\epsilon}$-smooth. We will then have the H-smooth integer $m_0 n_0$ lying in

$$x \ < \ mn \ < \ m_0 n_0 \ < \ (m+B)(n+B) \ < \ (m+B)(x/m+B) \ < \ x + 3B\sqrt{x}.$$

In total, then, we will have at least $\epsilon|N| > \epsilon^2\sqrt{x}/2$ products $m_0 n_0 \in [x, x+3B\sqrt{x}]$. Each m_0 and n_0 can each correspond to at most B different integers m, n; so, there are at least $\epsilon^2\sqrt{x}/(2B^2)$ *different* pairs (m_0, n_0) such that $m_0 n_0 \in [x, x+3B\sqrt{x}]$. Now, some of these pairs (m_0, n_0) will correspond to the same integer; that is, we may have $m_0 n_0 = m_1 n_1$. To handle the potential overcount, we note that each integer $j = m_0 n_0$ has fewer than $2\log\log y + 2(\log\log y)^{2/3}$ prime power factors, since m_0 and n_0 seperately have fewer than $\log\log y + (\log\log y)^{2/3}$ prime power factors. Thus, the number of divisors of j is

$$\tau(j) \ < \ 2^{2(\log\log y)+2(\log\log y)^{2/3}} \ < \ \log^{\log(4)+o(1)} y \ = \ \log^{\log(4)+o(1)} x.$$

Since each of our numbers j have so few possible factorizations, we must have that there are at least $\epsilon^2\sqrt{x}\log^{-\log(4)+o(1)}(x)/(2B^2)$ distinct integers $j \in [x, x+3B\sqrt{x}]$, all of whose prime divisors are at most H. The main theorem is then proved (upon dividing our ϵ by a factor of 20).

So we are left to show (1). First, define $f_y(n)$ to be the number of divisors of n of the form $p_1 p_2 p_3 p_4$, where the $p_i \in I := [y^{1/7-2\epsilon}, y^{1/7-\epsilon}]$ are distinct primes. Each $n \le y$ can have at most 7 such prime divisors and so we have that $f_y(n) \le \binom{7}{4}$ for each such n. In the remainder of this section we will show how (1) follows from the following proposition:

Proposition 1 *There exists a constant $\kappa > 0$ so that: If $0 < \epsilon < 1/200$, and if D is the set of all divisors $d = p_1 p_2 p_3 p_4$, where*

$$y^{1/7-2\epsilon} < p_1 < \cdots < p_4 < y^{1/7-\epsilon}$$

are primes, then,

$$\sum_{n\in[y,2y]} \left(\sum_{j=1}^{B} f_y(n+j) \ - \ B\overline{f_y} \right)^2 \ < \ \kappa B y, \tag{2}$$

where

$$\overline{f} \ = \ \overline{f}_y \ = \ \sum_{d\in D} \frac{1}{d}.$$

The proof of this proposition appears in the next section.

Now let $f^*(n) = f(n)g(n)$, where $g(n)$ is 1 if n has $< \log\log y + (\log\log y)^{2/3}$ prime power factors, and is 0 if n has more than this many prime power factors. A standard fact is that there are fewer than $|N|/\log\log y$ integers in $n \in N$ with $g(n) = 0$.

We claim that there exists $B_0 > 0$ so that if $B > B_0$, and x (and therefore y) is sufficiently large, there are at least $(1 - \epsilon/3)|N|$ integers $n \in N$ such that

$$\left| \sum_{j=1}^{B} f^*(n+j) \ - \ B\overline{f} \right| \ < \ \epsilon \overline{f} B. \tag{3}$$

Since $f(n) = f^*(n)$ for all but $O(|N|/\log\log y)$ elements of $n \in N$, this statement would follow if we could show that there exists $B_0 > 0$ so that for x sufficiently large and $B > B_0$, at least $(1 - \epsilon/4)|N|$ integers $n \in N$ satisfy

$$\left| \sum_{j=1}^{B} f(n+j) \ - \ B\overline{f} \right| \ < \ \epsilon \overline{f} B.$$

To see that such a $B_0 > 0$ exists, we observe that if we had more than $\epsilon|N|/4$ of the integers $n \in N$ fail to satisfy this inequality, then we would have

$$\sum_{n \in [y,2y]} \left| \sum_{j=1}^{B} f(n+j) \ - \ B\overline{f}_y \right|^2 \ \geq \ \epsilon^3 \overline{f}^2 B^2 (y_1 - y_0 - 2) \ > \ \epsilon^4 \overline{f}^2 B^2 y/4.$$

This cannot hold if $B > B_0 = 4\kappa\epsilon^{-4}\overline{f}^{-2}$, since it would contradict our proposition above. It remains to show that this B_0 is independent of y (that is, B_0 only depends of ϵ). To show this, we derive a formula for $\overline{f}$, and show that it only depends on ϵ, once y is sufficiently large:

$$\begin{aligned} \overline{f} \ &= \ \sum_{d \in D} \frac{1}{d} \ = \ \frac{1}{4!} \left(\sum_{\substack{y^{1/7-2\epsilon} < p < y^{1/7-\epsilon} \\ p \text{ prime}}} \frac{1}{p} \right)^4 \ - \ O\left(\sum_{n > y^{1/7}} \sum_{m < y} \frac{1}{mn^2} \right) \\ &= \ \frac{1}{4!} \log^4 \left(\frac{1/7 - \epsilon}{1/7 - 2\epsilon} \right) \ + \ O\left(\frac{1}{\log y} \right). \end{aligned}$$

We have used here the fact that $\sum_{p \leq t} 1/p \ = \ \log\log(t) + C + O(1/\log t)$, where the sum is over the primes and C is a constant; we have also used the fact that

$\sum_{n>t} 1/n^2 = O(1/t)$. We conclude that there are $(1-\epsilon/3)|N|$ integers $n \in N$ for which (3) holds.

Next, let $f'(n) = f^*(n)h(n) = f(n)g(n)h(n)$, where $h(n) = 1$ if n is $y^{3/(7\sqrt{e})+20\epsilon}$-smooth, and is 0 otherwise. We have that

$$\begin{aligned}
\sum_{n\in N} f'(n) &= \sum_{n\in N} f(n)h(n) \;+\; O\left(\frac{|N|}{\log\log|N|}\right) \;=\; \sum_{d\in D}\sum_{\substack{n\in N\\ d|n}} h(n) \;+\; O\left(\frac{|N|}{\log\log|N|}\right) \\
&= \sum_{d\in D} \#\{n\in[y_0/d, y_1/d] \;:\; n \text{ is } y^{3/(7\sqrt{e})+20\epsilon}-\text{smooth}\} \\
&\qquad + O\left(\frac{|N|}{\log\log|N|}\right) \\
&\geq \sum_{d\in D} \#\{n\in[y_0/d, y_1/d] \;:\; n \text{ is } (y_1/d)^{1/\sqrt{e}+7\epsilon}-\text{smooth}\} \\
&\qquad + O\left(\frac{|N|}{\log\log|N|}\right) \\
&= \sum_{d\in D} \frac{y_1-y_0}{d} - \#\{n\in[y_0/d, y_1/d] \;:\; \exists p|n,\; p \text{ prime},\; p > (y_1/d)^{1/\sqrt{e}+7\epsilon}\} \\
&\qquad + O\left(\frac{|N|}{\log\log|N|}\right) \\
&= \sum_{d\in D}\left(\frac{y_1-y_0}{d} - \sum_{\substack{(y_1/d)^{1/\sqrt{e}+7\epsilon}<p<y_1/d\\ p \text{ prime}}} \frac{y_1-y_0}{pd}\right) \;+\; O\left(\frac{|N|}{\log\log|N|}\right). \\
&= |N|\left(1+\log\left(\frac{1}{\sqrt{e}}+7\epsilon\right)\right)\overline{f} \;+\; O\left(\frac{|N|}{\log\log|N|}\right) \\
&> |N|\left(\frac{1}{2}+4\epsilon\right)\overline{f} \;+\; O\left(\frac{|N|}{\log\log|N|}\right).
\end{aligned}$$

This obviously implies

$$\sum_{n\in N}\sum_{j=1}^{B} f'(n+j) \;>\; B|N|\left(\frac{1}{2}+4\epsilon\right)\overline{f} + O\left(\frac{|N|}{\log\log|N|}\right).$$

Now, if (1) fails to hold, then there are at least $(1/2-2\epsilon)|N|$ integers $n \in N$ such that if $h \in [n+1, n+B]$ is an integer, then $f'(h) = 0$. Among these expectional n's,

at least $(1/2 - 7\epsilon/3)|N|$ of them satisfy (3). This then says that there exists a subset N_0 of N, having $|N_0| > (1/2 - 7\epsilon/3)|N|$ such that

$$\sum_{n \in N_0} \sum_{j=1}^{B} (f^*(n) - f'(n)) \ > \ B\overline{f}|N_0|(1-\epsilon) \ > \ B\overline{f}\left(\frac{1}{2} - \frac{10\epsilon}{3}\right).$$

This is impossible, however, because it would imply

$$\begin{aligned} B\left(1 + \frac{2\epsilon}{3}\right)|N|\overline{f} \ &< \ \sum_{n \in N} \sum_{j=1}^{B} f'(n) \ + \ \sum_{n \in N_0} \sum_{j=1}^{B} (f^*(n) - f'(n)) \\ &\leq \ B \sum_{n \in N} f^*(n) \ < \ B \sum_{d \in D} \frac{|N|}{d} \ + \ O(B|D|) \\ &< \ B|N|\overline{f} + O(B|N|^{4/7}). \end{aligned}$$

Thus, (1) holds, and the main theorem is proved.

3 Proof of Proposition 1

We have that

$$\sum_{n \in [y,2y]} \left(\sum_{j=1}^{B} f_y(n+j) \ - \ B\overline{f} \right)^2 \ = \ 2\Sigma_1 \ + \ \Sigma_2 \ - \ 2B\overline{f}\Sigma_3 \ + \ yB^2\overline{f}^2 \ + \ O(B^2),$$

where

$$\begin{aligned} \Sigma_1 \ &= \ \sum_{1 \leq j_1 < j_2 \leq B} \sum_{n \in [y,2y]} f_y(n+j_1) f_y(n+j_2); \\ \Sigma_2 \ &= \ \sum_{1 \leq j \leq B} \sum_{n \in [y,2y]} f_y(n+j)^2; \text{ and} \\ \Sigma_3 \ &= \ \sum_{1 \leq j \leq B} \sum_{n \in [y,2y]} f_y(n+j). \end{aligned}$$

For Σ_2 all we need is an upper bound: Since each $n < 2y + B$ can have at most 7 prime divisors $\geq y^{1/7-2\epsilon}$ (recall: $\epsilon < 1/200$) we have that $f(n+j) \leq \binom{7}{4}$ for $n \leq 2y$ and $j \leq B$. Thus, $\Sigma_2 \leq By\binom{7}{4}^2$.

For Σ_3 we get

$$\begin{aligned}\Sigma_3 &= B\sum_{n\in[y,2y]} f(n) \;+\; O(B^2) \;=\; B\sum_{d\in D}\#\{n\in[y,2y] \;:\; d|n\} \;+\; O(B^2)\\ &= By\sum_{d\in D}\frac{1}{d} \;+\; O(B|D|+B^2) \;=\; By\overline{f} \;+\; O(By^{4/7}).\end{aligned}$$

Finally, we estimate Σ_1, which is the most difficult of the three. Let $J=\lfloor y^\epsilon\rfloor+1$, and set $\alpha = y^{4\epsilon/J}$; note here that $\alpha-1$ has size about $(4\epsilon\log y)/y^\epsilon$. Next, for $j=1,2,...,J$ we define D_j to be the set of all $d\in D$ satisfying

$$y^{4/7-8\epsilon}\alpha^{j-1} \;\le\; d \;<\; y^{4/7-8\epsilon}\alpha^j.$$

Since $\alpha^J = y^{4\epsilon}$, and since $D\subset[y^{4/7-8\epsilon}, y^{4/7-4\epsilon}]$, one sees that the sets $D_1, D_2, ..., D_J$ partition D. Now, rewriting Σ_1 slightly we get

$$\begin{aligned}\Sigma_1 &= \sum_{j=1}^{B-1}\#\{j_1,j_2 \;:\; 1\le j_1<j_2\le B,\; j_2-j_1=j\}\sum_{n\in[y,2y]} f(n)f(n+j)\\ &\qquad + O(B^4)\\ &= \sum_{j=1}^{B-1}(B-j)\sum_{n\in[y,2y]} f(n)f(n+j) \;+\; O(B^4)\\ &= \sum_{j=1}^{B-1}(B-j)\sum_{d\in D}\sum_{e\in D}\sum_{\substack{m\in[y/d,2y/d]\\ e|dm+j}} 1 \;+\; O(B^4)\\ &= \sum_{j=1}^{B-1}(B-j)\sum_{k=1}^{J}\sum_{d\in D_k}\sum_{e\in D}\sum_{\substack{m\in[y/d,2y/d]\\ e|dm+j}} 1 \;+\; O(B^4)\end{aligned}$$

We now replace the inner sum with the sum over

$$m \;\in\; I_k \;:=\; \left[y^{3/7+8\epsilon}\alpha^{-k},\; 2y^{3/7+8\epsilon}\alpha^{-k+1}\right],\; e|dm+j,$$

and in so doing the interval that m lies in no longer depends on d. However, this introduces a small error, because this new interval is not exactly the same as $[y/d, 2y/d]$. Since all we need is an upper bound on Σ_1, we will not estimate this error explicitly, but will instead use the fact that $[y/d,2y/d]\subseteq I_k$ for $d\in D_k$ to deduce

$$\begin{aligned}\Sigma_1 \;\le\; \sum_{j=1}^{B-1}(B-j)\sum_{k=1}^{J}\sum_{e\in D}\#\{m\in I_k \;:\; -j\equiv dm \pmod e,\\ \text{for some } d\in D_k,\; (d,e)=1\} \;+\; O(B^4).\end{aligned} \tag{4}$$

The method we will use to bound this right-most expression from above using (4) will involve an application of bilinear Kloosterman sums estimates, and with some forethought one realizes that one needs to break up the expression into even more pieces in order to make such a method work: First, let A be the set of integers of the form p_1p_2, where $y^{1/7-2\epsilon} < p_1 < p_2 < y^{1/7-\epsilon}$ are prime, and for each $u \in A$, let $C_k(u)$ be the set of integers of the form p_3p_4 such that $y^{1/7-2\epsilon} < p_3 < p_4 < y^{1/7-\epsilon}$, and $p_1p_2p_3p_4 \in D_k$; that is, so that

$$p_3p_4 \in \left[\frac{y^{4/7-8\epsilon}\alpha^{k-1}}{p_1p_2},\ \frac{y^{4/7-8\epsilon}\alpha^{k}}{p_1p_2}\right),\ (p_3p_4, p_1p_2) = 1.$$

Then, we have that

$$\begin{aligned}\Sigma_1 \quad \leq \quad & \frac{1}{6}\sum_{j=1}^{B-1}(B-j)\sum_{k=1}^{J}\sum_{u\in A}\sum_{v\in C_k(u)}\sum_{\substack{e\in D\\(e,uv)=1}} \\ & \#\{m \in I_k \ :\ -j\overline{v}_e \equiv um \pmod{e}\}, \end{aligned} \tag{5}$$

where $\overline{v}_e$ is the multiplicative inverse of v modulo e; so, $v\overline{v}_e \equiv 1 \pmod{e}$. The reason for the factor $1/6$ is that there are 6 different ways of expressing an integer $d \in D$ as $d = uv$, where $u \in A$ and $v \in C_k(u)$.

Now, let

$$E(a,e,k,u) \ = \ \sum_{\substack{v\in C_k(u)\\(v,e)=1}} e\left(\frac{\overline{v}_e a}{e}\right),$$

and

$$F(a,e,k) \ = \ \sum_{m\in I_k} e\left(\frac{ma}{e}\right),$$

where $e(t)$ denotes $\exp(2\pi it)$. Then, using the fact that

$$\frac{1}{e}\sum_{a=1}^{e-1} e\left(\frac{-ja\overline{v}_e}{e}\right) e\left(\frac{-uma}{e}\right) \ = \ \begin{cases} 1, & \text{if } -j\overline{v}_e \equiv um \pmod{e};\ \text{and} \\ 0, & \text{if } -j\overline{v}_e \not\equiv um \pmod{e}, \end{cases}$$

one can rexpress (5) as $\Sigma_1 \leq (S_1 + S_2)/6 + O(B^4)$, where

$$S_1 \quad = \quad \sum_{j=1}^{B-1}(B-j)\sum_{k=1}^{J}\sum_{u\in A}\sum_{e\in D}\frac{E(0,e,k,u)F(0,e,k)}{e};\ \text{and}$$

$$\begin{aligned} S_2 &= \sum_{j=1}^{B-1}(B-j)\sum_{k=1}^{J}\sum_{u\in A}\sum_{e\in D}\frac{1}{e}\sum_{a\in[1,e-1]}E(-ja,e,k,u)F(-au,e,k) \\ &= \sum_{j=1}^{B-1}(B-j)\sum_{k=1}^{J}\sum_{\ell=1}^{J}\sum_{1\le a\le y^{4/7-4\epsilon}}\sum_{u\in A}\sum_{\substack{e\in D_\ell \\ e>a,\ (e,uv)=1}}\frac{1}{e}E(-ja,e,k,u)F(-au,e,k) \end{aligned}$$

It will turn out that S_1 gives the main contribution to Σ_1, and S_2 will be an error term.

We now estimte S_1: First, we note that for each $d \in D_k$, $|I_k| = (1+O(y^{-\epsilon}\log y))y/d$; and so, since $F(0,e,k)$ is the number of elements of I_k that are coprime to e, of which there are $|I_k|(1-O(y^{-\epsilon}))$ such elements, we deduce

$$F(0,e,k) \ = \ |I_k| \ = \ \frac{y}{|D_k|}\left(\sum_{d\in D_k}\frac{1}{d}\right)(1+O(y^{-\epsilon})).$$

So,

$$\begin{aligned} &\sum_{k=1}^{J}\sum_{u\in A}\sum_{e\in D}\frac{E(0,e,k,u)F(0,e,k)}{e} \\ &= \sum_{k=1}^{J}\frac{y}{|D_k|}\left(\sum_{d\in D_k}\frac{1}{d}\right)(1+O(y^{-\epsilon}\log y))\sum_{e\in D}\frac{\#\{uv\in D_k \ : \ u\in A,\ (uv,e)=1\}}{e} \\ &= \sum_{k=1}^{J}\frac{y}{|D_k|}\left(\sum_{d\in D_k}\frac{1}{d}\right)(1+O(y^{-\epsilon}\log y))\sum_{e\in D}\frac{6|D_k|}{e} \\ &= 6y(1+O(y^{-\epsilon}\log y)\left(\sum_{e\in D}\frac{1}{e}\right)\sum_{k=1}^{J}\sum_{d\in D_k}\frac{1}{d} \\ &= 6y\overline{f}^2(1+O(y^{-\epsilon}); \end{aligned}$$

and therefore,

$$S_1 \ = \ 3yB(B-1)\overline{f}^2(1+O(y^{-\epsilon}\log y).$$

In order to bound S_2 from above we require the following basic result from the theory of bilinear Kloosterman sums from [2] (Theorem 5, together with an adjustment, which is mentioned below).

Proposition 2 *Given a sequence of complex numbers β_n defined for integers $n \in (N, 2N]$, and given integers h and b, let*

$$\mathscr{C}(M, N; \beta) \;=\; \sum_{M<m\le 2M} \left| \sum_{\substack{N<n\le 2N \\ (n,m)=1}} \beta_n \; e\left(b\frac{\overline{n}}{m}\right)\right|^2 .$$

Then we have for any integer $0 < |b| \ll M$ that

$$\mathscr{C}(M, N; \beta) \;\ll\; ||\beta||^2 \tau(b)(M + N^2)N^{\epsilon},$$

where $||\beta||$ denotes the ℓ^2 norm of the vector $(\beta_{N+1}, ..., \beta_{2N})$.

Note: The version of this theorem appearing in the Duke et al. paper has $e(b\overline{m}/n)$, in place of $e(b\overline{n}/m)$. These two are related through the simple reciprocity relation

$$\frac{b\overline{m}}{n} \;\equiv\; \frac{-b\overline{n}}{m} \;+\; \frac{b}{mn} \pmod 1.$$

Thus,

$$e\left(\frac{b\overline{m}}{n}\right) \;=\; e\left(\frac{-b\overline{n}}{m}\right) \;+\; O\left(\frac{b}{MN}\right);$$

and so, with a little work, one can see that as long as $0 < |b| \ll M$, making the switch does not affect the upper bound presented above.

We will also require the following basic lemma.

Lemma 1 *For $-1/2 < a/e \le 1/2$, $a/e \ne 0$ we have that*

$$|F(a, e, k)| \;\le\; \frac{1}{2 < a/e >},$$

where $< t >$ denotes the distance from t to the nearest integer.

Proof of the Lemma. Since $F(a, e, k)$ is a geometric series, on applying the formula for the sum of the series we get

$$|F(a, e, k)| \;\le\; \frac{2}{|1 - e(a/e)|} \;=\; \frac{2}{|e(-a/(2e)) - e(a/(2e))|} \;=\; \frac{1}{2|\sin(\pi a/(2e))|}.$$

Then, applying the standard bound $|\sin(\pi t)| \geq 2|t|$ for $-1/2 \leq t \leq 1/2$ to this last term the lemma follows. ■

Now, fix $1 \leq j \leq B-1$, $1 \leq k \leq J$, $1 \leq \ell \leq J$, and $1 \leq a \leq y^{4/7-4\epsilon}$, and consider the inner two sums over u and e in our expansion for S_2 listed above: We have, on applying the Cauchy-Schwarz inequality and Lemma 1 that this sum is bounded from above by

$$\sum_{u\in A}\sum_{\substack{e\in D_\ell\\ e>a}} \frac{1}{e}|E(-ja,e,k,u)|\ |F(-au,e,k)|$$

$$\leq \sum_{u\in A}\left(\sum_{\substack{e\in D_\ell\\ e>a}}|E(-ja,e,k,u)|^2\right)^{1/2}\left(\sum_{\substack{e\in D_\ell\\ e>a}}\frac{1}{e}|F(-au,e,k)|^2\right)^{1/2}. \qquad (6)$$

To bound the first factor for the sum over $u \in A$, we apply Proposition 2 with parameters

$$b=-ja,\ N=\frac{y^{4/7-8\epsilon}\alpha^{k-1}}{u},\ M=y^{4/7-8\epsilon}\alpha^{\ell-1},$$

and where β_n is the indicator function of the set $C_k(u)$; that is, $\beta_n = 1$ if $n \in C_k(u)$, and equals 0 otherwise. With these choices of parameters we get

$$\sum_{\substack{e\in D_\ell\\ e>a}}|E(-ja,e,k,u)|^2 \ =\ \mathscr{C}(M,N;\beta) \ \ll\ |C_k(u)|\tau(a)\left(y^{4/7-8\epsilon}\alpha^\ell+\frac{y^{8/7-16\epsilon}\alpha^{2k}}{u^2}\right)y^\epsilon.$$

and so, the proposition gives

$$\begin{aligned}\sum_{\substack{e\in D_\ell\\ e>a}}|E(-ja,e,k,u)| &\ll \sqrt{|C_k(u)|}\tau(a)\left(y^{4/7-7\epsilon}\alpha^\ell+\frac{y^{6/7-11\epsilon}\alpha^{k+\ell/2}}{u}\right)\\ &\ll \tau(a)y^{5/7-4\epsilon}.\end{aligned}$$

Thus,

$$\begin{aligned}|S_2| &\ll B^2y^{5/7-4\epsilon}\sum_{k=1}^{J}\sum_{\ell=1}^{J}\sum_{u\in A}\sum_{1\leq a\leq y^{4/7-4\epsilon}}\frac{\tau(a)}{a} \ \ll\ B^2y^{1-4\epsilon}\sum_{a\leq y}\frac{\tau(a)}{a}\\ &\ll B^2y^{1-4\epsilon}\log^2(y),\end{aligned}$$

where this last inequality follows from the standard estimate

$$\begin{aligned}\sum_{a\le y}\frac{\tau(a)}{a} &= \sum_{1\le d\le y}\sum_{1\le n\le y/d}\frac{1}{dn}\\ &\ll \sum_{d\le y}\frac{1}{d}\sum_{n\le y}\frac{1}{n} \ll \log(y)\sum_{d\le y}\frac{1}{d} \ll \log^2(y).\end{aligned}$$

Thus, we have shown that S_2 is much smaller than S_1; and, in particular,

$$\Sigma_1 \le \frac{S_1+S_2}{6}+O(B^4) = \frac{1}{2}yB(B-1)\overline{f}^2(1+O(y^{-\epsilon}))+O(B^2).$$

From this is follows that

$$\begin{aligned}\sum_{n\in[y,2y]}\left(\sum_{j=1}^{B}f(n+j) - B\overline{f}\right)^2 &= 2\Sigma_1 + \Sigma_2 - 2B\overline{f}\Sigma_3 + yB^2\overline{f}^2\\ &= O(yB),\end{aligned}$$

and so the proposition is proved. ■

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